

CS471 – Web Technologies

Lab 1

The Internet Protocols

Prepared by: Jana Alrishan

Student ID: 421203667

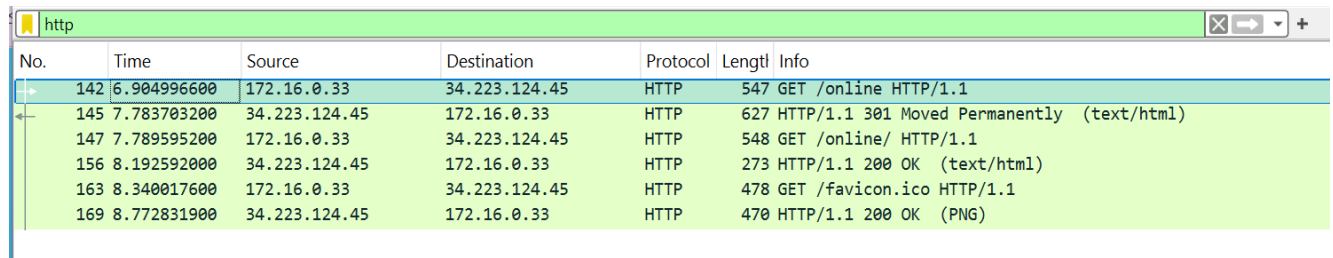
Section: 5527

Part 1: Capturing HTTP Traffic

Objective: Capture and analyze HTTP traffic using Wireshark.

Procedure: Wireshark was used to capture network traffic, then the http filter was applied to display HTTP packets.

HTTP Traffic Capture



The image shows a Wireshark packet capture window with the filter 'http' applied. The packet list displays six HTTP packets. Packets 142, 147, 156, and 163 are requests from 172.16.0.33 to 34.223.124.45. Packets 145 and 169 are responses from 34.223.124.45 to 172.16.0.33. The status bar at the bottom indicates 6 packets displayed (0-5).

No.	Time	Source	Destination	Protocol	Length	Info
142	6.904996600	172.16.0.33	34.223.124.45	HTTP	547	GET /online HTTP/1.1
145	7.783703200	34.223.124.45	172.16.0.33	HTTP	627	HTTP/1.1 301 Moved Permanently (text/html)
147	7.789595200	172.16.0.33	34.223.124.45	HTTP	548	GET /online/ HTTP/1.1
156	8.192592000	34.223.124.45	172.16.0.33	HTTP	273	HTTP/1.1 200 OK (text/html)
163	8.340017600	172.16.0.33	34.223.124.45	HTTP	478	GET /favicon.ico HTTP/1.1
169	8.772831900	34.223.124.45	172.16.0.33	HTTP	470	HTTP/1.1 200 OK (PNG)

Figure 1: Captured HTTP Traffic

The captured packets show HTTP requests and responses exchanged between the client and the server.

Part 2: Analyzing TCP/IP Traffic

Objective: Analyze the TCP connection, including connection establishment, data transfer, and connection termination.

2.1 Follow TCP Stream

Procedure: The TCP stream was followed to view the complete HTTP communication between the client and the server.

```
Wireshark · Follow TCP Stream (tcp.stream eq 6) · Wi-Fi

GET /online HTTP/1.1
Host: coolshiningoldlight.neverssl.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Referer: http://neverssl.com/
Accept-Encoding: gzip, deflate
Accept-Language: ar,en-US;q=0.9,en;q=0.8

HTTP/1.1 301 Moved Permanently
Date: Mon, 21 Sep 2026 04:25:55 GMT
Server: Apache/2.4.68 ()
Location: http://coolshiningoldlight.neverssl.com/online/
Content-Length: 295
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>301 Moved Permanently</title>
</head><body>
<h1>Moved Permanently</h1>
<p>The document has moved <a href="http://coolshiningoldlight.neverssl.com/online/">here</a>.</p>
</body></html>
```

Figure 2: TCP Stream (HTTP Request and Response)

The stream shows the HTTP GET request and the server response, including the 301 Moved Permanently response.

2.2 TCP Three-Way Handshake

Procedure: The TCP packets were examined to identify the three-way handshake used to establish the connection.

tcp.stream eq 6							
No.	Time	Source	Destination	Protocol	Length	Info	
43	3.085559400	172.16.0.33	34.223.124.45	TCP	66	57193 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2	
113	4.086555500	172.16.0.33	34.223.124.45	TCP	66	[TCP Retransmission] 57193 → 80 [SYN] Seq=0 Win=6553	
135	6.099850800	172.16.0.33	34.223.124.45	TCP	66	[TCP Retransmission] 57193 → 80 [SYN] Seq=0 Win=6553	
139	6.904296500	34.223.124.45	172.16.0.33	TCP	58	80 → 57193 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MS	
141	6.904497600	172.16.0.33	34.223.124.45	TCP	54	57193 → 80 [ACK] Seq=1 Ack=1 Win=65535 Len=0	

Figure 3: TCP Three-Way Handshake

The packets show the TCP connection establishment using SYN, SYN-ACK, and ACK.

2.3 TCP Packet Analysis

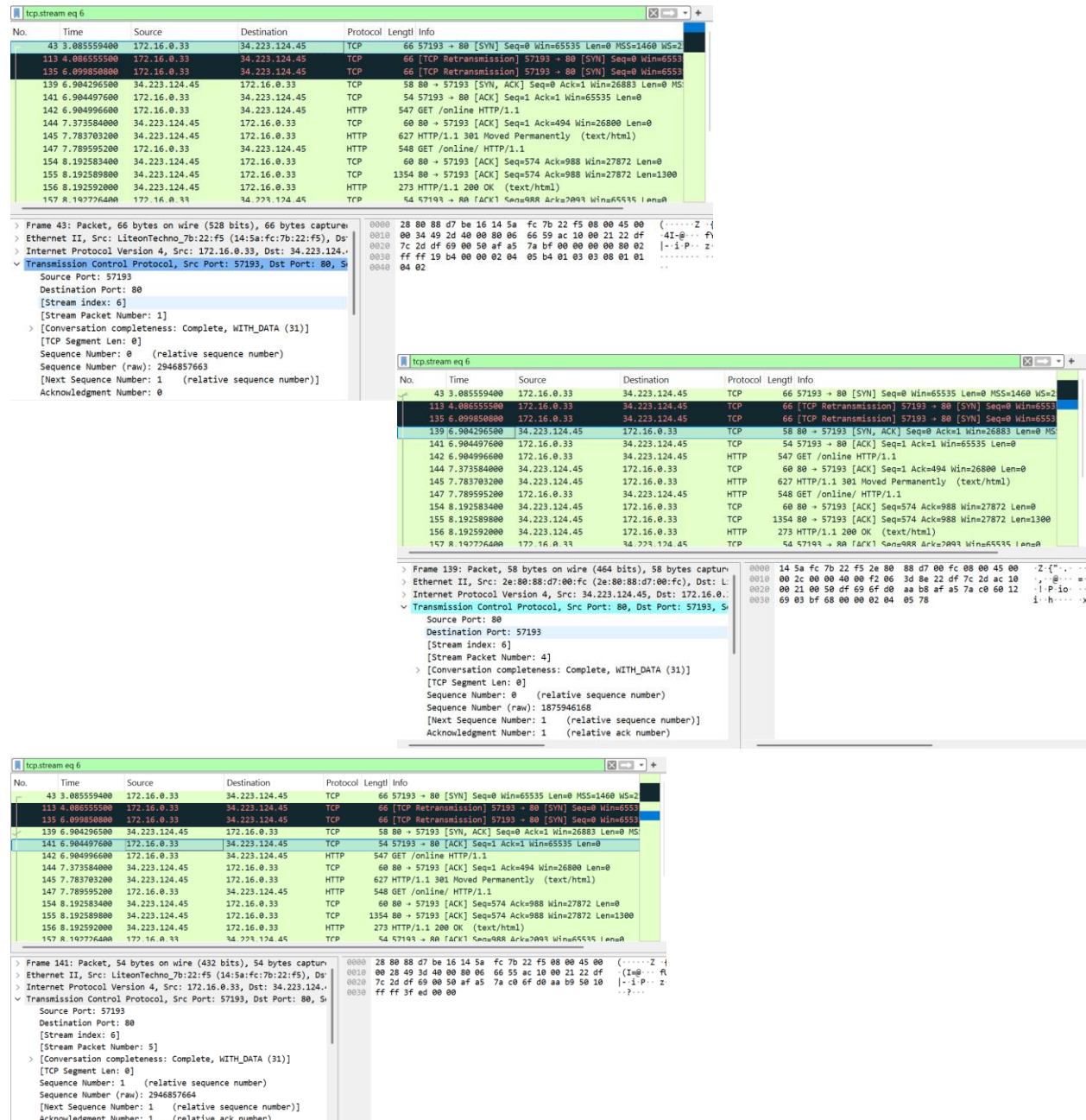


Figure 4: TCP Packet Details – Connection Establishment

The selected packets show TCP information such as source and destination ports, sequence numbers, and acknowledgment numbers during connection establishment.

2.4 TCP Data Transfer

No.	Time	Source	Destination	Protocol	Length	Info
43	3.085559400	172.16.0.33	34.223.124.45	TCP	66	57193 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2
113	4.086555500	172.16.0.33	34.223.124.45	TCP	66	[TCP Retransmission] 57193 → 80 [SYN] Seq=0 Win=65535
135	6.099850800	172.16.0.33	34.223.124.45	TCP	66	[TCP Retransmission] 57193 → 80 [SYN] Seq=0 Win=65535
139	6.904296500	34.223.124.45	172.16.0.33	TCP	58	80 → 57193 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS=1460
141	6.904497600	172.16.0.33	34.223.124.45	TCP	54	57193 → 80 [ACK] Seq=1 Ack=1 Win=65535 Len=0
142	6.904996600	172.16.0.33	34.223.124.45	HTTP	547	GET /online HTTP/1.1
144	7.373584000	34.223.124.45	172.16.0.33	TCP	60	80 → 57193 [ACK] Seq=1 Ack=494 Win=26800 Len=0
145	7.783703200	34.223.124.45	172.16.0.33	HTTP	627	HTTP/1.1 301 Moved Permanently (text/html)
147	7.789595200	172.16.0.33	34.223.124.45	HTTP	548	GET /online/ HTTP/1.1
154	8.192583400	34.223.124.45	172.16.0.33	TCP	60	80 → 57193 [ACK] Seq=574 Ack=988 Win=27872 Len=0
155	8.192589800	34.223.124.45	172.16.0.33	TCP	1354	80 → 57193 [ACK] Seq=574 Ack=988 Win=27872 Len=1300
156	8.192592000	34.223.124.45	172.16.0.33	HTTP	273	HTTP/1.1 200 OK (text/html)
157	8.192726400	172.16.0.33	34.223.124.45	TCP	54	57193 → 80 [ACK] Seq=988 Ack=2093 Win=65535 Len=0

> Frame 155: Packet, 1354 bytes on wire (10832 bits), 1354 bytes captured (10832 bits) on interface 0

> Ethernet II, Src: 2e:80:88:d7:00:fc (2e:80:88:d7:00:fc), Dst: 172.16.0.33

> Internet Protocol Version 4, Src: 34.223.124.45, Dst: 172.16.0.33

> Transmission Control Protocol, Src Port: 80, Dst Port: 57193, Seq: 574, Len: 1300

Source Port: 80

Destination Port: 57193

[Stream index: 6]

[Stream Packet Number: 11]

> [Conversation completeness: Complete, WITH_DATA (31)]

[TCP Segment Len: 1300]

Sequence Number: 574 (relative sequence number)

Sequence Number (raw): 1875946742

[Next Sequence Number: 1874 (relative sequence number)]

Acknowledgment Number: 988 (relative ack number)

0000 14 5a fc 7b 22 f5 2e 80 88 d7 00 fc 08 00 45 00 :Z-{"...
0010 05 3c 69 96 40 00 f2 06 ce e7 22 df 7c 2d ac 10 :<i@...
0020 00 21 00 50 df 69 6f d0 ac f6 af a5 7e 9b 50 10 :!Pio...
0030 6c e0 53 c4 00 00 48 54 54 50 2f 31 2e 31 20 32 :1S...HT
0040 30 30 20 4f 4b 0d 0a 44 61 74 65 3a 20 4d 6f 6e :00OK...D
0050 2c 20 32 31 20 53 65 70 20 32 30 32 36 20 30 34 : , 21 Sep
0060 3a 32 35 3a 35 36 20 47 4d 54 0d 0a 53 65 72 76 : :25:56 G
0070 65 72 3a 20 41 70 61 63 68 65 2f 32 2e 34 2e 36 :er: Apac
0080 38 20 28 29 0d 0a 4c 61 73 74 2d 4d 6f 64 69 66 :8 ()...La
0090 69 65 64 3a 20 57 65 64 2c 20 32 39 20 4a 75 6e :ied: Wed
00a0 20 32 30 32 32 20 30 30 3a 32 33 3a 32 32 20 47 : 2022 00
00b0 4d 54 0d 0a 45 54 61 67 3a 20 22 38 62 65 2d 35 :MT...ETag
00c0 65 32 38 62 32 39 32 39 31 65 31 30 2d 67 7a 69 :e28b2929
00d0 70 22 0d 0a 41 63 63 65 70 74 2d 52 61 6e 67 65 :p"...Acce
00e0 73 3a 20 62 79 74 65 73 0d 0a 56 61 72 79 3a 20 :s: bytes
00f0 41 63 63 65 70 74 2d 45 6e 63 6f 64 69 6e 67 6d :Accept-E
0100 0a 43 6f 6e 74 65 6e 74 2d 45 6e 63 6f 64 69 6e :Content

Figure 5: TCP Data Transfer

The selected packet shows data being transferred from the server to the client as part of the HTTP communication.

2.5 TCP Connection Termination

No.	Time	Source	Destination	Protocol	Length	Info
144	7.373584000	34.223.124.45	172.16.0.33	TCP	60	80 → 57193 [ACK] Seq=1 Ack=494 Win=26800 Len=0
145	7.783703200	34.223.124.45	172.16.0.33	HTTP	627	HTTP/1.1 301 Moved Permanently (text/html)
147	7.789595200	172.16.0.33	34.223.124.45	HTTP	548	GET /online/ HTTP/1.1
154	8.192583400	34.223.124.45	172.16.0.33	TCP	60	80 → 57193 [ACK] Seq=574 Ack=988 Win=27872 Len=0
155	8.192589800	34.223.124.45	172.16.0.33	TCP	1354	80 → 57193 [ACK] Seq=574 Ack=988 Win=27872 Len=1300
156	8.192592000	34.223.124.45	172.16.0.33	HTTP	273	HTTP/1.1 200 OK (text/html)
157	8.192726400	172.16.0.33	34.223.124.45	TCP	54	57193 → 80 [ACK] Seq=988 Ack=2093 Win=65535 Len=0
163	8.340017600	172.16.0.33	34.223.124.45	HTTP	478	GET /favicon.ico HTTP/1.1
169	8.772831900	34.223.124.45	172.16.0.33	HTTP	470	HTTP/1.1 200 OK (PNG)
170	8.820674100	172.16.0.33	34.223.124.45	TCP	54	57193 → 80 [ACK] Seq=1412 Ack=2509 Win=65119 Len=0
194	13.934604400	34.223.124.45	172.16.0.33	TCP	54	80 → 57193 [FIN, ACK] Seq=2509 Ack=1412 Win=28944 Len=0
195	13.934675500	172.16.0.33	34.223.124.45	TCP	54	57193 → 80 [ACK] Seq=1412 Ack=2510 Win=65119 Len=0

> Frame 194: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface 0

> Ethernet II, Src: Netgear_d7:be:16 (28:80:88:d7:be:16), Dst: 172.16.0.33

> Internet Protocol Version 4, Src: 34.223.124.45, Dst: 172.16.0.33

> Transmission Control Protocol, Src Port: 80, Dst Port: 57193, Seq: 2509, Len: 0

Source Port: 80

Destination Port: 57193

[Stream index: 6]

[Stream Packet Number: 17]

> [Conversation completeness: Complete, WITH_DATA (31)]

[TCP Segment Len: 0]

Sequence Number: 2509 (relative sequence number)

Sequence Number (raw): 1875948677

[Next Sequence Number: 2510 (relative sequence number)]

Acknowledgment Number: 1412 (relative ack number)

0000 14 5a fc 7b 22 f5 2e 80 88 d7 be 16 08 00 45 00 :Z-{"...
0010 00 28 69 99 40 00 f2 06 d3 f8 22 df 7c 2d ac 10 :<i@...
0020 00 21 00 50 df 69 6f d0 b4 85 af a5 80 43 50 11 :!Pio...
0030 71 10 bf 8c 00 00 :q.....

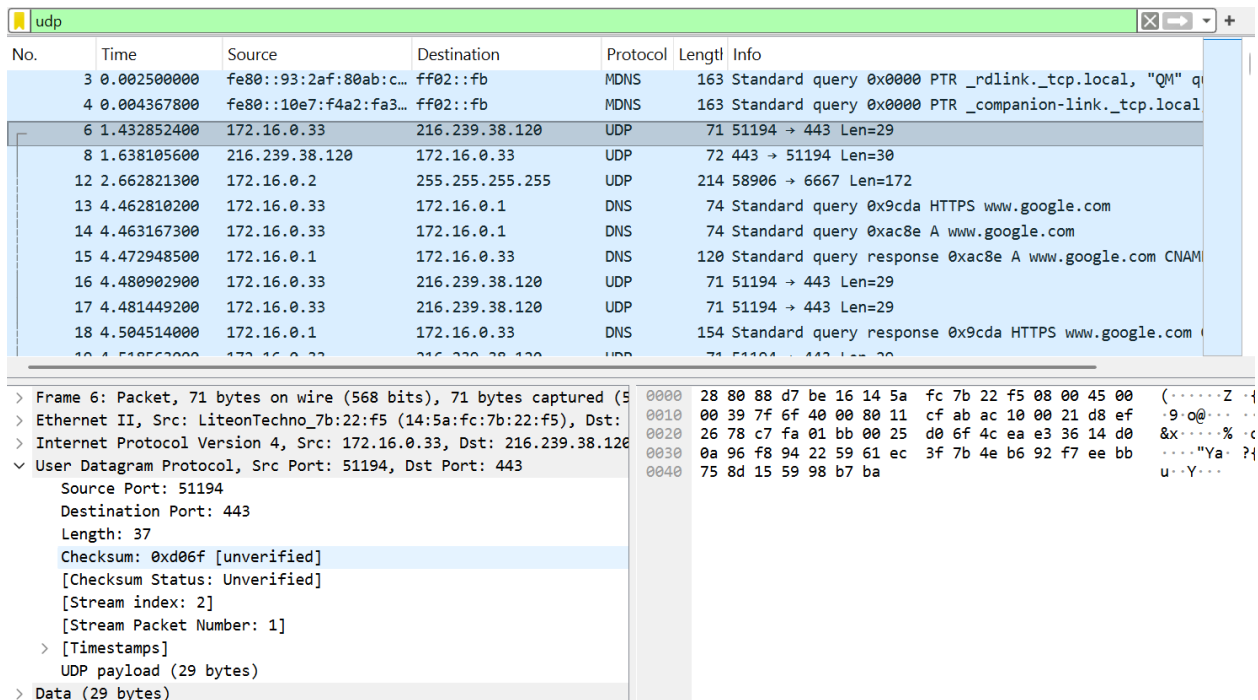
Figure 6: TCP Connection Termination

The FIN, ACK packet indicates the termination of the TCP connection, followed by the final acknowledgment.

Part 3: Capturing UDP Traffic

Objective: Capture and analyze UDP traffic using Wireshark.

Procedure: The udp filter was applied to display UDP packets, and one packet was selected to inspect its header information.



No.	Time	Source	Destination	Protocol	Length	Info
3	0.002500000	fe80::93:2af:80ab:c...	ff02::fb	MDNS	163	Standard query 0x0000 PTR _rdlink._tcp.local, "QM" q
4	0.004367800	fe80::10e7:f4a2:fa3...	ff02::fb	MDNS	163	Standard query 0x0000 PTR _companion-link._tcp.local
6	1.432852400	172.16.0.33	216.239.38.120	UDP	71	51194 → 443 Len=29
8	1.638105600	216.239.38.120	172.16.0.33	UDP	72	443 → 51194 Len=30
12	2.662821300	172.16.0.2	255.255.255.255	UDP	214	58906 → 6667 Len=172
13	4.462810200	172.16.0.33	172.16.0.1	DNS	74	Standard query 0x9cda HTTPS www.google.com
14	4.463167300	172.16.0.33	172.16.0.1	DNS	74	Standard query 0xac8e A www.google.com
15	4.472948500	172.16.0.1	172.16.0.33	DNS	120	Standard query response 0xac8e A www.google.com CNAM
16	4.480902900	172.16.0.33	216.239.38.120	UDP	71	51194 → 443 Len=29
17	4.481449200	172.16.0.33	216.239.38.120	UDP	71	51194 → 443 Len=29
18	4.504514000	172.16.0.1	172.16.0.33	DNS	154	Standard query response 0x9cda HTTPS www.google.com

Frame 6: Packet, 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on interface 0	
Ethernet II, Src: LiteonTechno_7b:22:f5 (14:5a:fc:7b:22:f5), Dst: 01:00:0c:00:00:00	28 80 88 d7 be 16 14 5a fc 7b 22 f5 08 00 45 00 (.....Z..)
Internet Protocol Version 4, Src: 172.16.0.33, Dst: 216.239.38.120	00 39 7f 6f 40 00 80 11 cf ab ac 10 00 21 d8 ef .9.o@....
User Datagram Protocol, Src Port: 51194, Dst Port: 443	00 20 26 78 c7 fa 01 bb 00 25 d0 6f 4c ea e3 36 14 d0 &x.....%..c
Source Port: 51194	00 30 0a 96 f8 94 22 59 61 ec 3f 7b 4e b6 92 f7 ee bb"Ya. ?{
Destination Port: 443	00 40 75 8d 15 59 98 b7 ba u...Y...
Length: 37	
Checksum: 0xd06f [unverified]	
[Checksum Status: Unverified]	
[Stream index: 2]	
[Stream Packet Number: 1]	
[Timestamps]	
UDP payload (29 bytes)	
Data (29 bytes)	

Figure 7: UDP Traffic and Packet Details

The captured UDP traffic shows communication without a connection establishment process. The selected packet displays information such as the source port, destination port, length, and UDP payload.

Comparison Between TCP and UDP

Task 1

	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	Reliable and uses a three-way handshake to establish a connection.
Data Integrity and Ordering	TCP	Ensures data arrives correctly and in the correct order.

Task 2

	TCP	UDP
Use cases	Web browsing, email, file transfer	Streaming, online games, DNS
Performance	Slower due to connection setup and reliability mechanisms	Faster with lower overhead and no connection setup

Conclusion

In this lab, HTTP traffic was captured and analyzed using Wireshark. TCP connection establishment and termination were examined, as well as UDP traffic. The analysis showed the main differences between TCP and UDP in terms of reliability, connection establishment, and performance.